Raspberry Pi Foundation

Lesson 4: Bias in, bias out

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If you are an educator, ask your students to complete a short survey: rpf.io/exai-st.

Learners will explore how bias can appear in machine learning models due to the data used to train them. They will create their own machine learning model to classify images of apples and tomatoes, and discover how a limited dataset can lead to biased and inaccurate predictions. Finally, they will investigate two types of bias that can appear in training data.

Objectives

Objective 1	Describe the impact of data on the accuracy of a machine learning (ML) model
Objective 2	Explain the need for both training and test data
Objective 3	Explain how bias can influence the predictions generated by an ML model

Keywords

- Training data
- Test data
- Accuracy
- Bias
- Data bias
- Societal bias

Lesson outline

Activity	Objective	Description	Time
Activity 1: Data types for classification		Learners identify different data types that can be used for classification.	10 mins
Activity 2: Supermarket AI application	1, 2	Learners create an ML model using Machine Learning for Kids and discuss the importance of using representative datasets.	20 mins
Activity 3: Bias	3	Learners identify data bias and societal bias as two types of bias in training data and discuss ways they can reduce them.	10 mins
Optional: Your future career	1, 3	Learners apply their understanding of bias to a career prediction scenario.	~15 mins
Exit ticket	1, 2, 3	Learners review key concepts about bias by answering one short question.	5 mins

Resources

Computers with internet access

Printed/Digitally distributed
- Worksheet 2: Supermarket AI application
- Worksheet (optional): Your future career

Links
- machinelearningforkids.co.uk
- ai-activities.raspberrypi.org/project-files
- [rpf.io/xai-3-v1](rpf.io/xai-3-v1)

Subject knowledge

Activity 1: You will need to be able to support learners in identifying what type of data would be used for the example classification problems, as well as which classes could be used. Answers are provided in the slide deck.

Activity 2: Learners will use Machine Learning for Kids (machinelearningforkids.co.uk) as a tool to create a machine learning model. Following step-by-step instructions in their worksheets, they will create two classes and label training data, then train and test their model. Before you deliver the lesson, we recommend that you watch the educator support video (rpf.io/ml4kids-demo) provided, so that you are familiar with the steps involved.

Activity 3: The focus of this activity is the concept of bias and how it appears in machine learning models. The slides are scaffolded to introduce data bias and societal bias. Encourage discussion from the learners around the examples. The final slide in the activity states that “Societal bias can lead to data bias”. The example on the slide — a facial recognition system that is less accurate in recognising people with certain skin tones — is an example of data bias that can be caused by societal bias: this data bias can occur when not enough data from a range of people with different skin tones is used to train the model, which can happen due to societal bias. More detail about bias in machine learning is provided in the lesson walkthrough.

Assessment opportunities

Objective	Description
1	Activity 2: You can assess learners’ ability to describe the impact of data on the accuracy of a machine learning model through the discussion questions at the end of this activity. Exit ticket: You can assess learners’ answers to the first question.
2	Activity 2: You can assess learners’ ability to explain the need for both training and test data through the discussion questions at the end of this activity. Exit ticket: You can assess learners’ answers to the second question.
3	Activity 3: You can assess learners’ understanding of the key differences between data bias and societal bias during discussions.

Lesson walkthrough

The **Success criteria** on slide 6 translate lesson objectives into actionable, student friendly targets. They clearly define what the students should be able to do by the end of the lesson.

Slide deck labels appear at the top of each slide to help you navigate the activities in the lesson.

This lesson includes an **optional activity** that will take around 15 minutes. If you do not want to include it, please skip slides 23–27. These slides have been hidden by default.

10 mins	**Activity 1: Data types for classification**
Slide 2	Ask learners to choose one of the three example classification problems on the slide and think about what type of data they would use and what some of the classes could be. Highlight that almost any type of data can be used for a classification system, not just images or text.
	Educator guidance If learners cannot remember what a class is, instruct them to look at the **student glossary** for a reminder. Use the **'think, pair, share' pedagogy** for this activity: once learners have chosen one of the classification examples, ask them to **think** about what type of data they would use and what some of the classes could be. Then, instruct them to **pair** up with their neighbour, or someone who has chosen a different classification example, to discuss their answers. Then, ask some of the learners to **share** their answers with the class. Before you move on to slides 3–5 to reveal the answers, ask a few learners to explain why they would use the data types and classes they have chosen. When discussing sound data, you could ask learners what file types this might involve, for example, WAV and MP3.
Slides 3–5	Use the slide animations to reveal the answers.
20 mins	**Activity 2: Supermarket AI application**
Slide 7	Introduce the scenario for this activity: a supermarket would like to use the cameras around its stores to identify the items customers have placed in their baskets. Stress the word "cameras" to the learners, as it is important that they notice how the data will be collected.
Slide 8	Highlight to the learners that they will be required to create a machine learning model. Ask the learners, "What is a data-driven model?" This is to remind learners of the concepts covered in Lesson 2. Give learners time to discuss their answers in pairs, then ask the class for their answers. Use the animation to reveal the answer.
	Educator guidance To recap, a data-driven model is a representation of a real-world context. ML developers use vast amounts of data that is representative of a specific problem to train a model to detect patterns. The result of the training is a model, which is used to produce predictions about new data in the same context.
Slide 9	Tell the learners that they are going to create their own data-driven model to identify apples and tomatoes. Ask learners what type of data will be collected from the cameras.

Give learners time to discuss their answers in pairs, then ask the class for their answers.

If learners answer "data", ask them specifically what type. The answer is that they will need **images** of apples and tomatoes — and lots of them.

Slide 10

Use the animation on this slide to describe how data is used to create a supervised learning model.

Educator guidance

When training a machine learning model, it is important that the data is separated into 'training data' and 'test data'. The model is trained using the training data.

After that, the accuracy of the model is tested using the test data, which has not been used as part of the training process. Once the model has been trained and tested, it is ready to be used in the real world.

Slide 11

Resource required: Worksheet 2: Supermarket AI application

Explain to learners that they are going to create their own machine learning model that classifies tomatoes and apples. Before they begin, give a quick demonstration of the process using Machine Learning for Kids (machinelearningforkids.co.uk). Walk through the steps they will need to follow on the worksheet.

After your demonstration, hand out **Worksheet 2: Supermarket AI** and ask learners to follow the instructions to train their model. Allow 5 minutes for learners to complete the training process.

Educator guidance

Before the lesson, we recommend that you watch the educator support video (rpf.io/ml4kids-demo) so that you are familiar with the steps involved before you demonstrate the activity.

Students do not have to create an account to complete this project.

The learners will use the worksheet (which includes all the steps to complete the activity) and Machine Learning for Kids for this activity. They will need to complete the following tasks:

- Create a project that recognises images
- Create a label (class) for apples and one for tomatoes
- Add training data (5 images of apples and 5 images of tomatoes) from the webpage linked on the worksheet (ai-activities.raspberrypi.org/project-files) — encourage the learners to think about which images they want to use to train the model
- Train the model

Slide 12

Explain that learners now need to test their model using the test data provided. Allow 3 minutes for testing. Encourage the learners to test their model with as many images from the test data provided as possible.

Educator guidance

The learners should find that their model's predictions are not as accurate as they need to be. They can see this by looking at what label the model has applied to the test data,

	and looking at the confidence score.
Slide 13	Ask the learners the questions on the slide to encourage them to think about why the model's predictions were not accurate enough for the supermarket. Instruct them to think, pair, share to answer the questions.

	Educator guidance
	
	The answers that the learners give should be around the two main issues:
	1. Not enough data was used to train the model.
	2. The 'right' data was not used:
	    a. There were no images of red apples in the training data.
	    b. The images were not very representative of how the apples and tomatoes might look if they were on a supermarket shelf. For example, the training data included images of apples hanging from trees.

10 mins	**Activity 3: Bias**
Slide 14	Read the questions on the slide to the learners, and explain that they are going to watch Stevie Bergman, a research scientist at Google DeepMind, explain what bias is. Play the video (3 minutes). After the video, ask the learners to answer the questions on the slide.
	Educator guidance The answers are as follows: 1. Bias in ML is when the model produces an output that treats one thing, person, or group differently to another, either positively or negatively. 2. Bias can come in at almost any point during the development of an ML model, including when developing the problem statement, during data collection, and when evaluating the model. 3. There are many different kinds of biases, including bias in data (data bias) and when the model's training data reflects a bias that exists in society (societal bias). The focus of Activity 3 is to introduce learners to how machine learning models can produce biased predictions. The activity includes opportunities for discussion. Try to keep to time to allow enough time to complete the activity.
Slide 15	Use this slide to help students reflect on the questions posed beside the video. Ask for a volunteer to answer each question in turn and use the animation to reveal the answer.
	Educator guidance An alternative way to deliver this activity would be to print this slide without the answers and ask students to fill in the answers on the sheet. You could also pause the video when the answer to one of the questions has been discussed and ask the class what the answer was.
Slide 16	Describe what bias is in a machine learning context.

	Read the two examples of machine learning bias on the slide, but at this stage, do not discuss them or how they could have happened.
Slides 17–18	Describe **data bias**. Ask the learners to recall how data bias appeared in the model they made for the supermarket. Display the next slide to show several questions they can think about to help reduce data bias as much as possible.
Slide 19	Describe **societal bias**. Read the example on the slide and ask the learners to think about how societal bias could appear in the data used to create the fictional AI application. Take suggestions from the class.
	Educator guidance There are many possible answers to this question, for example: • Jobs that are less common now, like typist or bank teller, could appear more common • The dataset is unlikely to reflect the diversity of the modern workforce in the country in which the data was collected • Certain genders could be more likely to be associated with specific jobs (this varies between countries)
Slide 20	Ask the learners which gender might have been associated with each job on the slide in 1960. Ask whether or not this societal bias could lead to the AI application producing biased predictions.
	Educator guidance If there is time, you could highlight this point further by demonstrating an image search (using a search engine of your choice) for the jobs on the slide. You could ask learners, “What if those images were used as part of the training data? What impact would this have on the predictions produced by a machine learning model?”
Slides 21–22	Use the slides to assess learners’ understanding of the difference between data bias and societal bias. Click through the slides to revisit the examples given at the start of this activity and ask learners to vote on whether each one is an example of societal bias or data bias. Use the slide animations to reveal the answers.
	Educator guidance Some learners might think that a facial recognition system that is less accurate in recognising people with certain skin tones, is an example of societal bias. However, it is an example of data bias that can be caused by societal bias: the model has likely not been trained with enough data from a range of people with different skin tones, which can happen due to societal bias.
15 mins	**Optional activity: Your future career** **Resource required: Worksheet: Your future career**
	Educator guidance This is an optional extension activity. If you do not want to include it, please skip slides 23–27. **Alternative approach**

You could use **Worksheet: Your future career** to capture your learners' thoughts, but if you are unable to print the worksheet, you could show the questions on the slides for each task and ask learners to write their answers on a sheet of paper instead.

Slide 23 **Optional**

Introduce the activity to your learners: your school is thinking of using an AI application to help predict which career learners might choose when they are older.

Slide 24 **Optional**

Reveal what the model has predicted will be the most appropriate career choices for some of your learners. You can point out that this is a completely fictional outcome, but hopefully it will cause light-hearted discontent among some of your learners. Ask the learners how the model could have produced those predictions.

Educator guidance

Before the lesson, on this slide, add the names of some of your learners to the jobs that a fictional machine learning model has predicted as their careers. This slide is deliberately provocative and is designed to cause debate about the accuracy of the model. It should prompt your learners to think about how the model produced the predictions it did (what data was used to train the model).

Slide 25 **Optional**

Ask the learners to complete the first task in **Worksheet: Your future career**, or answer the questions on the slide on a sheet of paper. Allow 5 minutes for the learners to complete this task. They need to evaluate the proposed sets of data and write down which ones they think could be suitable to be used in training the model, and which ones they think would be unsuitable. They must also choose one dataset they have decided would be unsuitable and explain why they have made this decision.

Educator guidance

The purpose of this activity is to encourage learners to think about how the suitability of a dataset will impact how effective a machine learning model is.

None of the datasets would be completely free from bias. Even if an up-to-date dataset were used, it would contain current societal biases. The focus of this activity is to encourage discussion and critical thinking, rather than to decide on the correct answer.

The data from the school would be the most appropriate in terms of its relevance to the learners, but the data that is 10 years old would be out of date. It would not include new types of jobs and it would reflect societal biases from the last 10 years.

The regional career and education data is more current, but might not be as relevant to the learners as it has been collected from a wider group.

National employment data from 1960 to 2000 would be out of date and would reflect societal biases from the time. It would not contain new types of jobs and would include jobs that do not exist any more. The data would also be too general and not as relevant.

Data from the international job market would form a very large dataset and would be very general and so would not be very useful.

Slide 26 **Optional**

Instruct learners to complete the second task on the worksheet, or answer the questions on the slide on a sheet of paper. They should look at the categories of data included in the datasets and decide which categories they would remove to try to reduce bias in the model's output. They must choose one category they think should be removed and

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	explain how it could introduce bias into the model if it was used for training.
Slide 27 **Optional**	Ask students to discuss in pairs which categories of data they would remove from the model and how one of these categories could introduce bias. Then, ask for learners to volunteer to share their thoughts.
5 mins	**Exit ticket**
Slides 28–29	Use the multiple choice question to ask learners how bias in machine learning can be reduced. Move on to the next slide to reveal the answer.
Slides 30–31	Use the multiple choice question to ask learners why we need training data. Move on to the next slide to reveal the answer.
Slides 32–33	Use the multiple choice question to ask learners why we need test data. Move on to the next slide to reveal the answer.
Slide 33	Use this slide to share a recap of Lessons 1–4.